



Confidence-Driven Region Refinement for Small Object Detection in UAV Images

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ABSTRACT

Small object detection in UAV imagery is challenging due to tiny targets and limited computing resources on edge devices. We propose a confidence-driven region adaptive reasoning method based on YOLOv5. The framework first performs fast global low-resolution detection to obtain rough results, then selects medium-confidence regions for local high-resolution fine detection, and finally fuses global and local results via NMS. Experiments on VisDrone show that our method improves mAP_{0.5} by 3.6% and increases FPS from 30 to 35 compared with YOLOv5, achieving better accuracy with higher speed.

Keywords: confidence-driven; region adaptive reasoning; YOLOv5; small object detection; UAV

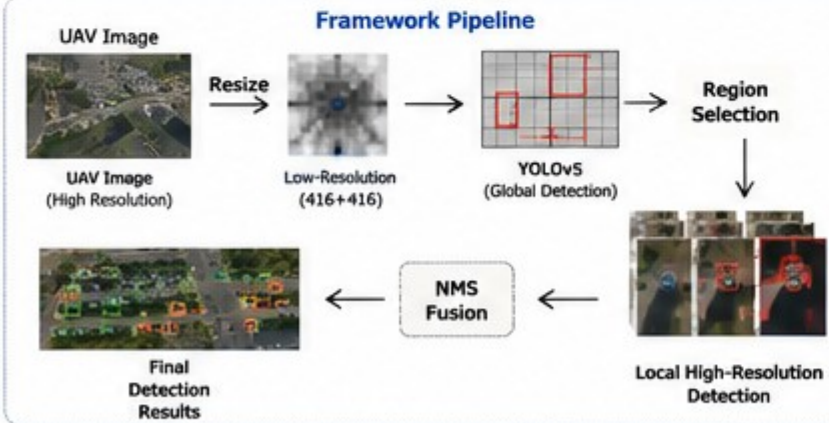
1. MOTIVATION

- UAV images contain numerous small targets with low pixel ratios and blurry details.
- Edge devices have limited computing power, constraining complex deep models.
- Existing methods either lose accuracy (low resolution) or cause high computation (dense slicing / full high-res inference).
- We propose a confidence-driven adaptive framework to balance accuracy and efficiency.



2. METHOD OVERVIEW

The proposed framework consists of four modules.



7. EVALUATION METRICS & DATASET



Dataset: VisDrone2019-det

• Train: 6,471 images • Val: 548 images • Test: 1,610 images

Metrics:

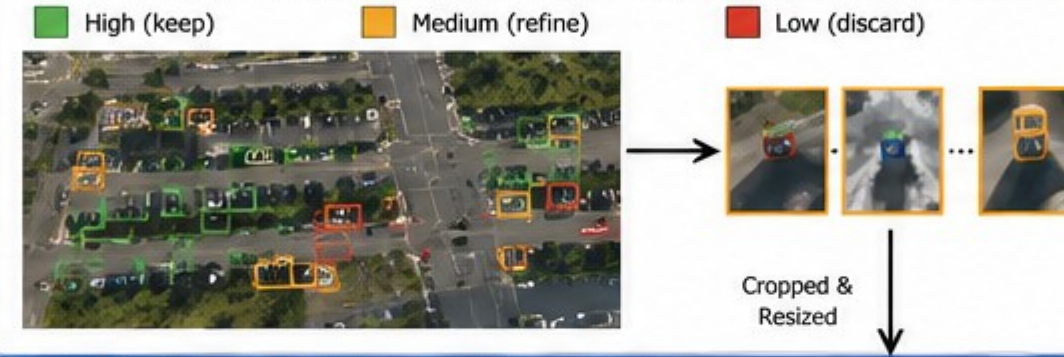
• mAP_{0.5} • mAP_{0.5:0.95} (COCO-style)
• FPS (frames per second)

3. CONFIDENCE-DRIVEN REGION SELECTION

Let the global detection results be $B = [b_i, c_i]$, $i = 1, 2, \dots, n$, where b_i is the i -th bounding box and $c_i \in [0, 1]$ is its confidence score. We set low and high thresholds T_l and T_h ($T_l < T_h$). Regions are divided into three groups:

- High-confidence (keep): $c_i \geq T_h \rightarrow$ Output directly
- Medium-confidence (refine): $T_l \leq c_i < T_h \rightarrow$ Enter local high-resolution detection
- Low-confidence (discard): $c_i < T_l \rightarrow$ Discard

$$B = \{b_i \mid T_l \leq c_i < T_h\} \quad (1)$$



4. LOCAL HIGH-RESOLUTION FINE DETECTION

Each selected region is cropped with context, upscaled, and fed into YOLOv5 again.



5. RESULT FUSION (NMS)

Global results and local refined results are merged. Duplicate boxes are removed by Non-Maximum Suppression (NMS) based on IoU.

Intersection over Union (IoU) between two boxes B_i and B_j :

$$IoU(B_i, B_j) = \frac{Area(B_i \cap B_j)}{Area(B_i \cup B_j)} \quad (2)$$

If $IoU > \tau$ (e.g., 0.5), keep the box with higher confidence.



OUR CONTRIBUTIONS

- ✓ Propose a confidence-driven region adaptive reasoning framework for UAV small object detection.
- ✓ Introduce dynamic region selection to allocate high-resolution computation only to informative regions.
- ✓ Achieve higher detection accuracy (+3.6% mAP_{0.5}) and faster inference (30 $\rightarrow$ 35 FPS) on VisDrone.
- ✓ Provide an efficient and deployable solution for edge devices with limited resources.

8. CONCLUSION



We propose a confidence-driven region adaptive YOLOv5 framework that adaptively refines medium-confidence regions with high-resolution inference. Experiments on VisDrone demonstrate that our method significantly improves detection accuracy and speed, making it well suited for real-time UAV applications on resource-constrained edge devices.

6. EXPERIMENTS AND RESULTS

6.1 Overall Performance Comparison

Method	mAP _{0.5} (%) $\uparrow$	mAP _{0.5:0.95} (%) $\uparrow$	FPS (frame/s) $\uparrow$
YOLOv5 (Baseline)	68.5	38.5	30
Ours (Proposed)	72.1	41.3	35

6.2 Ablation Study

Method (Add one by one)	mAP _{0.5} (%) $\uparrow$	FPS (frame/s) $\uparrow$
YOLOv5 (Baseline)	68.5	30
+ Low-resolution global detection	69.4	38
+ Confidence region screening	70.3	37
+ Local high-resolution fine detection (Ours)	72.1	35

6.3 Parameter Sensitivity (Thresholds)

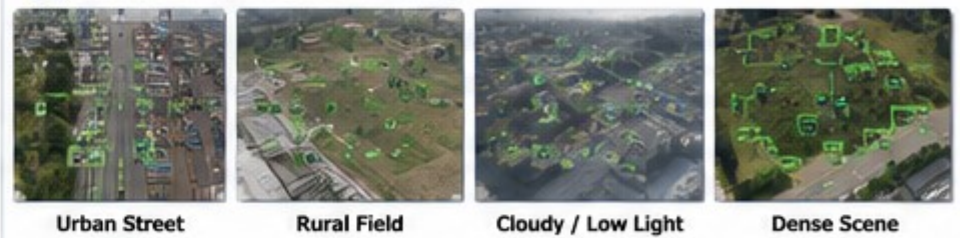
Low/High Threshold (T_l/T_h)	mAP _{0.5} (%) $\uparrow$	FPS (frame/s) $\uparrow$
0.2 / 0.6	71.8	32
0.3 / 0.7	72.1	35
0.4 / 0.8	72.4	28

6.4 Qualitative Results

(a) Comparison with YOLOv5 (Same Scene)



(b) Ours Results in Different Environments



9. REFERENCES

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